A direct conditional comparison up to $ns = 3/100$

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28 October 2026

Prepared with AI assistance. This note proves a conditional comparison, not the full Berry–Esseen conjecture.

1 Keeping the third-moment gain extends the direct comparison

The direct argument can be extended from $ns \le 1/50$ to $ns \le 3/100$ without a parameter grid. The useful change is to keep the favorable third-moment term: it pays for most of the normal variance correction. The negative and positive inner phases must also have different incoming tail bounds. The support assumption remains 3/5 throughout.

Theorem 1. *Let $n \ge 42$, $2/5 \le p \le 53/100$, $q = 1 - p$, and $Y = B + U$, with $\mathbb{P}(B = 1) = p$ and $\mathbb{E}(U \mid B) = 0$. For the two conditional laws assume*

$$|U_j| \le 3/5, \qquad U_0 \le a_0, \quad U_1 \ge -a_1, \qquad a_0, a_1 \ge 0, \quad a_0 + a_1 \le 9/125.$$

Set $v = pq$, $s = \mathbb{E}U^2$, $\lambda = ns$, $X = (Y - p)/\sqrt{v + s}$, $\rho = \mathbb{E}|X|^3$, and $\rho_B = (1 - 2v)/\sqrt{v}$. Suppose $0 < \lambda \le U_* := 3/100$, $|H| \le 46/25$, and $C \ge 2/5$. Choose an integer k and $u \in [-34/125, 91/125]$ such that $np + H\sqrt{v + s} = k + u$, and write $\delta = k - np$. Then k is interior, and for the sum S_n of n independent copies of X,

$$\begin{aligned}\Delta_C := \sqrt{n}\{\mathbb{P}(S_n \le H) - \Phi(H/\sqrt{n})\} - C\rho \\ - \sqrt{n}\{F_B(k) - \Phi(\delta/\sqrt{nv})\} + C\rho_B < -\lambda/100,\end{aligned}$$

where F_B is the distribution function of $\mathrm{Bin}(n, p)$.

This theorem deliberately requires $C \ge 2/5$. The preceding 1/50 theorem, which permits every $C \ge 0$, is unchanged. In particular the new theorem applies at the sharp Bernoulli constant. Acquiring the conditional representation, strip and threshold bound remains a separate part of the global proof.

1.1 The comparison and its repaired normalization

Let W_j be a sum of j independent U_1 variables and $n - j$ independent U_0 variables, and put $\pi_j = \binom{n}{j} p^j q^{n-j}$. Define

$$f(u) = \frac{\Phi((\delta + u)/\sqrt{n(v + s)}) - \Phi(\delta/\sqrt{n(v + s)})}{\pi_k},$$

$$O = \mathbb{P}(W_k > u) + f(u), \qquad T = \sum_{i=1}^{n-k} \mathbb{P}(W_{k+i} \le u - i).$$

Conditioning on the count and discarding only nonpositive outgoing terms other than the own count gives

$$\Delta_C \le -\sqrt{n}\pi_k(O - R_0 T) + \mathcal{R}, \tag{1}$$

provided $\pi_{k+i}/\pi_k \le R_0$, where

$$\mathcal{R} = \sqrt{n}\{\Phi(\delta/\sqrt{nv}) - \Phi(\delta/\sqrt{n(v+s)})\} + C(\rho_B - \rho).$$

The own outgoing event is strict; every incoming event is closed. This identity and convention include atom collisions.

Write $A = 3/5$, $v_* = 6/25$, and $\Xi = (46/25)(501/1000)$. As $\sqrt{v+s} < 501/1000$, $|\delta| < 33/20$ globally, while $|u| \le 3/40$ gives $|\delta| < \Xi + 3/40 < 1$. Thus $1 \le k \le n-1$ and $9/25 < k/n < 23/40$. Finite Stirling gives

$$\sqrt{n}\pi_k = \frac{\phi(0)}{\sqrt{v}} \exp\{-\delta^2/(2nv) + e\}, \qquad n|e| < Z(D) := 5D/8 + 3/8 + D^3/42$$

for $|\delta| \le D \le 33/20$. The proof in the preceding direct-comparison note uses the entropy, logarithmic-prefactor and factorial-remainder constants $6, 5/4, 3/8$ on $[9/25, 23/40]$; these constants and this segment are unchanged. In particular crossing $p = 1/2$ is allowed. For fixed actual k and δ,

$$\log f'(t) = -\tfrac{1}{2}\log(1 + s/v) + \frac{\delta^2}{2nv} - \frac{(\delta + t)^2}{2n(v+s)} - e.$$

Consequently, when $|t| \le b$,

$$n|\log f'(t)| \le \tfrac{8}{3}\lambda + \frac{2Db + b^2}{2v_*} + Z(D).$$

Take $(D, b) = (1, 3/40)$ in the inner interval. Using $e^{-x} \ge 1 - x$ and $e^x \le (1-x)^{-1}$ proves

	$\lvert u\rvert \le 3/40$	$-34/125 \le u \le 91/125$
$\sqrt{n}\pi_k$ lower bound	73/100	13/20
f' lower bound	24/25	4/5
f' upper bound	26/25	5/4
R_0 upper bound	53/50	3/2

The same finite estimates also give $f' > 19/20$ on $0 \le u \le 1/10$ and $f' > 9/10$ throughout the positive phase interval. For the former use $(D, b) = (21/20, 1/10)$; for the latter use the signed bound $2\delta t + t^2 \le \max(0, 2\Xi u - u^2)$, $0 \le t \le u$. The ratio arguments are unchanged: the binomial mode has $\sqrt{n}\pi_j < 17/20$; in the inner interval the first count ratio is less than $53/50$ and every later factor at most one because $(n+1)p - k < 153/100 < 2$. The exact checker rechecks all these finite normalization inequalities at the enlarged noise endpoint.

Here is the new gain. With $r_p = \min(p, q)$ and $d = a_0 + a_1$, conditional signs and cubic expansion give

$$\mathbb{E}|Y - p|^3 \ge v(1 - 2v) + (3r_p - d)s.$$

Using $(1+x)^{3/2} \le 1 + 3x/2 + 3x^2/8$, $r_p \ge 2/5$ and $v \ge 6/25$, we obtain

$$(v+s)^{3/2}(\rho - \rho_B) \ge s\left(\frac{87}{250} - \frac{13s}{16}\right) \ge sB_*, \qquad B_* := \frac{87}{250} - \frac{13U_*}{16 \cdot 42}. \tag{2}$$

The exact inequality

$$\frac{2}{5}B_* - (501/1000)^3 = \frac{92514493}{7000000000} > 0$$

therefore proves $C(\rho-\rho_B)>s=\lambda/n$. Differentiating the normal variance correction bounds it by $(7/4)\delta_+\lambda/n$. Thus

$$\mathcal{R}<\frac{3\lambda}{4n}\quad(|u|\le 3/40),\qquad \mathcal{R}<\frac{2\lambda}{n}\quad\text{globally.}\tag{3}$$

The $1/n$ factor is essential: the gain is relative to s, not to λ without normalization.

As before, $|s_1-s_0|\le(5/2)s$, where $s_j=\mathbb{E}U_j^2$. Hence, with $t=\mathbb{E}W_k^2$,

$$\frac{79}{84}\lambda\le t\le\frac{89}{84}\lambda\quad(|u|\le 3/40),\qquad t\le\frac{11}{10}\lambda\quad\text{globally.}$$

For $i\ge 1$, $\mathbb{E}W_{k+i}^2\le iV\lambda$, with $V=9/8$ in the inner interval and $V=7/6$ globally.

1.2 Incoming tails and the own count

The following incoming bound was proved in the 1/50 note from Pinelis, arXiv:0902.4058, Proposition 2.8 and Corollary 2.11. Put $\mu=V\lambda/A^2$ and

$$Q(\mu,h)=\frac{\mu^2(1/2+\mu/6)}{((1-h)/A-1+\mu)^2},$$
$$J(\mu,h)=\frac{\mu^3}{3(1/A-1-h/(2A)+\mu)^2(1-3\mu)}.$$

When $u\le h$, $h\ge 0$, $3\mu<1$, and $b_h:=1/A-1-h/(2A)>0$, J bounds the sum of incoming probabilities for $i\ge 2$. The unsquared hinge denominator is then positive. If also $h<1-A$, Q bounds the first incoming probability. These follow from the integer hinges $j=1$ and $j=i$, respectively, and $\mathbb{E}(P_\mu-i)_+^2\le\mu^{i+1}/(i+1)!$ for $i\ge 2$. The coefficient ratio in the resulting series is less than 3μ. Moreover, J/λ increases with λ; so does Q/λ if $\mu\le(1-h)/A-1$. These are precisely the hypotheses checked at U_* below. In the negative inner interval one uses $h=0$.

For the own count, set $\alpha=13/25$ and use

$$h_\alpha(x)=\frac{-x^4+3\alpha^2x^2+2\alpha^3x}{5\alpha^4}\le\mathbf{1}_{(0,\infty)}(x).$$

Its maximum on the positive half-line is $(9+6\sqrt{3})/20<1$; its numerator factors as $-x(x+\alpha)^2(x-2\alpha)$. For a centered independent sum with absolute summand bound A, $|\mathbb{E}W^3|\le At$ and $\mathbb{E}W^4\le A^2t+3t^2$. Writing $r=zt$, the shifted quartic plus the linear normal term, divided by t, is bounded below by

$$F_{\sigma,b}(z,t)=\frac{3\alpha^2-A^2-3t}{5\alpha^4}+\left(\frac{2\sigma}{5\alpha}+b-\frac{4At}{5\alpha^4}\right)z$$
$$+\frac{3\alpha^2t-6t^2}{5\alpha^4}z^2-\frac{t^3z^4}{5\alpha^4}.$$

Set $T_*=(89/84)U_*=89/2800$. Concavity in t reduces the rectangle $0\le t\le T_*$ to its two endpoints. On $0\le z\le 3/2$, replace z^4 by $(3/2)^2z^2$, then check the endpoints and any eligible quadratic vertex. The result is

$$F_{1,-26/25}>37/100,\qquad F_{-1,24/25}>19/20.$$

The exact positive margins are

$$\frac{82564663499}{16050459443200},\qquad\frac{377266419344095319}{17248259356596450000}.$$

For negative phase beyond $r = 3t/2$, Cantelli gives $O \geq r^2/(t + r^2) - (26/25)r$. Its derivative is positive at $r = 3t/2$ and first increases then decreases, so its minimum on $[3t/2, 3/40]$ is at an endpoint. Both endpoint values exceed $(37/100)t$, uniformly for $t \leq T_*$. For positive phase beyond $3t/2$, $O \geq (24/25)r > (19/20)t$. The switch lies below $3/40$. Thus

$$O > \frac{37}{100}\frac{79}{84}\lambda \quad (-3/40 \leq u \leq 0), \qquad O > \frac{19}{20}\frac{79}{84}\lambda \quad (0 \leq u \leq 3/40). \tag{4}$$

1.3 Six closed intervals, with no noise grid

All quantities now use $U_* = 3/100$. Write $\mu_I = 3/32$, $\mu_G = 7/72$, and $t_G = (7/6)U_*$. For each interval let B denote the following lower bound for $O - R_0T$, with Q, J evaluated at the displayed arguments:

$[-3/40, 0]$	$B/\lambda = (37/100)(79/84) - (53/50)[Q(\mu_I, 0) + J(\mu_I, 0)]/U_*$
$[0, 3/40]$	$B/\lambda = (19/20)(79/84) - (53/50)[Q(\mu_I, 3/40) + J(\mu_I, 3/40)]/U_*$

For the outer negative interval set

$$L_- := \min_{r \in \{3/40, 34/125\}} \{r^2/((11/10)U_* + r^2) - (5/4)r\}.$$

The derivative is positive at $3/40$ and can have only a later maximum, so $O \geq L_-$ throughout. There take $B = L_- - (3/2)[Q(\mu_G, 0) + J(\mu_G, 0)]$. For $[3/40, 1/10]$ take

$$B = \frac{19}{20}\frac{3}{40} - \frac{3}{2}\{t_G/(t_G + (9/10)^2) + J(\mu_G, 1/10)\}.$$

On each of $[1/10, 1/5]$ and $[1/5, 91/125]$, the function $(9/10)u - (3/2)t_G/(t_G + (1 - u)^2)$ is concave, because $3(1 - u)^2 > t_G$. Let its endpoint minimum be L_+ and let h be the interval's right endpoint. There take $B = L_+ - (3/2)J(\mu_G, h)$.

All six brackets are positive. After multiplying by the appropriate mass lower bound, subtracting (3), and allowing the target margin $\lambda/100$, the exact residuals are:

$[-3/40, 0]$ (divided by λ)	19650157/2820720000
$[0, 3/40]$ (divided by λ)	40612185886957/129085181232000
$[-34/125, -3/40]$	12343549039/533914920000
$[3/40, 1/10]$	1221121/371280000
$[1/10, 1/5]$	652372219/42906045000
$[1/5, 91/125]$	40879152773/857753190000

In the outer intervals, use U_* for all increasing errors and the decreasing own-count lower bound; therefore the displayed absolute positive residuals work for every smaller positive noise. In the inner intervals, monotonicity of $(Q + J)/\lambda$ gives a bound proportional to λ all the way to zero. The six intervals are closed, meet at their endpoints, and cover $[-34/125, 91/125]$. This proves Theorem 1.

The exact checker retains the preceding normalization gates and checks the new gates separately. Its negative controls remove or reverse the cubic gain, omit its $1/n$ scaling, reuse the inappropriate positive phase tail bound on negative phases, and enlarge the target to .04. These controls reject the sufficient estimates; they are not counterexamples to the Berry–Esseen conjecture. The full interval $.03 < \lambda \leq .075$ is still outside the proved theorem.